-------- SQL Practice for begginners----------

```
create table student (
student_id int primary key,
first_name varchar,
last_name varchar,
GPA float,
enrollment_date date time,
major varchar
);
Select * from student;
insert into student values (201,'Shivansh','Mahajan',8.79,'2021-09-01
9:30','Computer Science');
insert into student values (202,'Umesh','Sharma',8.44,'2021-09-01
08:30','Mathematics');
insert into student values (203,'Rakesh','Kumar',5.6,'2021-09-01 10:00','Biology');
insert into student values (204,'Radha','Sharma',9.2,'2021-09-01
12:45','Chemistry');
insert into student values (205,'Kush','Kumar',7.85,'2021-09-01 08:30','Physics');
insert into student values (206,'Prem','Chopra',9.56,'2021-09-01 9:24','History');
insert into student values (207,'Pankaj','Vats',7.78,'2021-09-01 02:30','English');
insert into student values (208,'Navleen','Kaur',7,'2021-09-01
06:30','Mathematics');

create table program(
student_ref_id int references student(student_id),
Program_name varchar,
program_start_date date
);
select * from program;

insert into program values (201,'Computer Science','2021-09-01');
insert into program values (202,'Mathematics','2021-09-01');
insert into program values (208,'Mathematics','2021-09-01');
insert into program values (205,'Physics','2021-09-01');
insert into program values (204,'Chemistry','2021-09-01');
insert into program values (207,'Psycology','2021-09-01');
insert into program values (206,'History','2021-09-01');
insert into program values (203,'Biology','2021-09-01');
```

```
create table scholarship(
student_ref_id int references student(student_id),
scholarship_amount int,
scholarship_date date
);

insert into scholarship values (201,5000,'2021-10-15');
insert into scholarship values (202,4500,'2022-08-18');
insert into scholarship values (203,3000,'2021-01-25');
insert into scholarship values (204,4000,'2021-10-15');
```

--1. Write a SQL query to fetch "FIRST_NAME" from the Student table in upper case and use ALIAS name as STUDENT_NAME.

```
select upper(first_name) student_name from student;
```

--2. Write a SQL query to fetch unique values of MAJOR Subjects from Student table.

```
select distinct(major) from student;
```

--3. Write a SQL query to print the first 3 characters of FIRST_NAME from Student table.

```
select substring(first_name,1,3) from student;
```

--4. Write a SQL query to find the position of alphabet ('a') in the first name column 'Shivansh' from Student table.

```
select position ('a' in first_name) from student where first_name ='Shivansh';
```

--5. Write a SQL query that fetches the unique values of MAJOR Subjects from Student table and print its length.

```
select distinct(major) as Majors,length(major) as major_length from student
order by majors;
```

--6. Write a SQL query to print FIRST_NAME from the Student table after replacing 'a' with 'A'.

```
select replace(first_name,'a','A') from student;
```

--7. Write a SQL query to print the FIRST_NAME and LAST_NAME from Student table into single column COMPLETE_NAME.

```
select (first_name || ' '||last_name) Full_Name from student;
select concat(first_name, ' ', last_name) as Full_Name from student;
```

--8. Write a SQL query to print all Student details from Student table order by FIRST_NAME Ascending and MAJOR Subject descending .

```
select * from student order by first_name, major desc;
```

--9. Write a SQL query to print details of the Students with the FIRST_NAME as 'Prem' and 'Shivansh' from Student table.

```
select * from student where first_name in ('Prem' ,'Shivansh');
select * from student where first_name = 'Prem' or first_name = 'Shivansh';
```

--10. Write a SQL query to print details of the Students excluding FIRST_NAME as 'Prem' and 'Shivansh' from Student table.

```
select * from student where first_name not in ('Prem' ,'Shivansh');
select * from student where first_name != 'Prem' and first_name != 'Shivansh';
```

--11. Write a SQL query to print details of the Students whose FIRST_NAME ends with 'a'.

```
select * from student where first_name like '%a';
```

--12. Write an SQL query to print details of the Students whose FIRST_NAME ends with ‘a’ and contains five alphabets.

```
select * from student where first_name like '____a';
```

--13. Write an SQL query to print details of the Students whose GPA lies between 9.00 and 9.99.

```
select * from student where gpa between 9.00 and 9.99;
```

--14. Write an SQL query to fetch the count of Students having Major Subject ‘Computer Science’.

```
select count(student_id) total_count from student where major = 'Computer Science';
```

--15. Write an SQL query to fetch Students full names with GPA >= 8.5 and <= 9.5.

```
select (first_name || ' '||last_name) Full_Name , gpa from student where gpa >= 8.5 and gpa <= 9.5;
```

--16. Write an SQL query to fetch the no. of Students for each MAJOR subject in the descending order.

```
select major, count(student_id) total_student from student group by major order by count(student_id) desc;
select *, count(major) over (partition by major) from student;
```

--17. Display the details of students who have received scholarships, including their names, scholarship amounts, and scholarship dates.

```
select (s.first_name || ' ' || s.last_name) as
Name,sc.scholarship_amount,sc.scholarship_date from student s
join
scholarship sc on s.student_id = sc.student_ref_id;
```

--18. Write an SQL query to show only odd rows from Student table.

```
select * from student where student_id % 2 != 0;
```

--19. Write an SQL query to show only even rows from Student table

```
select * from student where student_id % 2 = 0;
```

--20. List all students and their scholarship amounts if they have received any. If a student has not received a scholarship, display NULL for the scholarship details.

```
select * from student s
full join
scholarship sc on s.student_id = sc.student_ref_id;
```

--21. Write an SQL query to show the top n (say 5) records of Student table order by descending GPA.

```
select * from student order by gpa desc limit 5;
```

--22. Write an SQL query to determine the nth (say n=5) highest GPA from a table.

```
SELECT * FROM Student ORDER BY GPA DESC LIMIT 1 offset 4;
select *,
nth_value(gpa,5) over (order by gpa desc) from student limit 1 offset 4;
select * from student s
where 4 = (select count(distinct(s1.gpa))
from student s1 where s1.gpa > s.gpa
);
```

-- 25. Write an SQL query to show the second highest GPA from a Student table using sub-query.

```
select max(gpa) from student where gpa not in (select max(gpa) from student);

with ranked_gpa as (
        select *,
        dense_rank() over (order by gpa desc) as rank
```

```
        from student)
select * from ranked_gpa where rank = 2;
```

--27. Write an SQL query to list STUDENT_ID who does not get Scholarship.

```
select student_id from student where student_id not in (select student_ref_id
from scholarship);
select * from student s
left join scholarship sc
on s.student_id = sc.student_ref_id
where sc.student_ref_id is null;
```

--28. Write an SQL query to fetch the first 50% records from a table.

```
select * from student
limit (select count(*)/2 from student);
```

--29. Write an SQL query to fetch the MAJOR subject that have less than 4 people in it.

```
with count_major as (
select distinct(major),
count(major) over (partition by major) major_count from student)
select * from count_major where major_count < 4 order by major_count;
select major, count(major) from student group by major having count(major) < 4;
```

--31. Write an SQL query to show the last record from a table.

```
select * from student where student_id = (select max(student_id) from student);
```

--32. Write an SQL query to fetch the first row of a table.

```
select * from student order by student_id limit 1;
```

--33. Write an SQL query to fetch the last 50% records from a table.

```
select * from student
limit (select count(*)/2 from student)
offset (select count(*)/2 from student);
```

--39. Write an SQL query to show the current date and time

```
select current_date;
select current_date, current_time;
```

--41. Write an SQL query to update the GPA of all the students in 'Computer Science' MAJOR subject to 7.5.

```
update student
set gpa = 7.5 where major = 'Computer Science';
```

--45. Write an SQL query to find the students who have the same GPA as 'Shivansh Mahajan'

```
select * from student where gpa = (select gpa from student where student_id = 201);
select * from student where gpa = 7.5;
```